



A Case Study of Classroom Podcast in Ohlone College

Dr. Jim D. Pham

Ohlone College

jpham@ohlone.edu

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Agenda

- Background
- What is podcasting
- A Case Study of Classroom Podcast in Ohlone College
- Results of the Study
- Getting Started With Podcasting
- Podcasting Equipments
- Recommendations and Conclusions
- Q & A



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PodCast

Background

- Since the 1990s, Internet and Web technologies have created opportunities for exchanging information, technology-assisted teaching and learning, as well as online and distance learning for higher education.
- Like the Internet and Web technologies, emerging mobile and entertainment technologies add new dimensions for students to learn, communicate, and exchange information.
- The Pew Internet and American Life Project (2009) reported that 74% of teens owned an iPod or MP3 player and 60% of teens owned a personal computer. They also reported that mobile phone use had climbed from 63% in the 12- to 17-year-old group in fall 2006 to 71% in early 2008.



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What is Podcasting?

- The process of creating audio/video (or multimedia) file and distributing it on the Internet
- Audio/video podcast delivered by subscription through RSS (Really Simple Syndication)
- Audio/video podcast playbacks on any mobile devices (cell phone, PDA, etc.), MP3 (iPad, iPod, iTouch, etc.) devices or personal computers



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Why Classroom Podcast?

- Podcasting represents new thinking in the delivery of educational content for playback lectures and lessons on student mobile devices, MP3 players or personal entertainment devices.
- Podcasting allows faculty to create, publish, and distribute course content on the Internet.
- Podcasting allows students to access on demand the learning materials and download the podcasts either from faculty Web sites or from Apple's iTunes software application.



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Statement of the Problem

- Podcasting methods offers new flexible learning outside the classroom, but it also presents challenges to traditional classroom learning.
- One of the major challenges facing classroom learning is how to engage the diverse learning styles of student, especially community college students.
- The benefits of incorporating classroom podcasts into traditional classroom settings remain unclear.
- With increasing domination by mobile technologies, it seems that academia is trying to redefine its image beyond just the site for teaching and learning.



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A Case Study of Classroom Podcast in Ohlone College

This study examined the use of classroom podcasts as supplemental course materials for playback of lectures and other lessons and the relationship of this use with the test scores.

- (a) The study examined the effects of classroom podcasts and the use of mobile and portable media devices, and the impact of these devices on student learning.
- (b) The prediction was that students who listened to classroom podcasts were more likely to receive higher test scores than those who did not.



Research Question

Are there differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) with regard to the students' test scores?



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PodCast

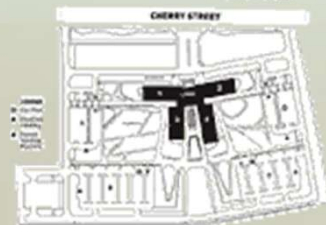
The Site for the Study

- Ohlone College is a part of the Ohlone Community College District, with campuses in Fremont and Newark.
- Ohlone College is located in the proximity of the Silicon Valley, best known for the silicon chip innovators and the headquarters of such leading high-technology manufacturers as Apple Computer, Ebay, Google, Hewlett Packard, Intel, Oracle, Sun, Yahoo, and many more.

OHLONE COLLEGE FREMONT
43600 Mission Boulevard
Fremont, CA 94539-5847
(916) 659-4000



OHLONE COLLEGE NEWARK
Center for Health Sciences and Technology
39399 Cherry Street
Newark, CA 94560-4902
(916) 742-2300



Ohlone College
43600 Mission Boulevard
Fremont, CA 94539-5847

Ohlone College Newark Center
for Health Sciences and Technology
39399 Cherry Street
Newark, CA 94560-4902



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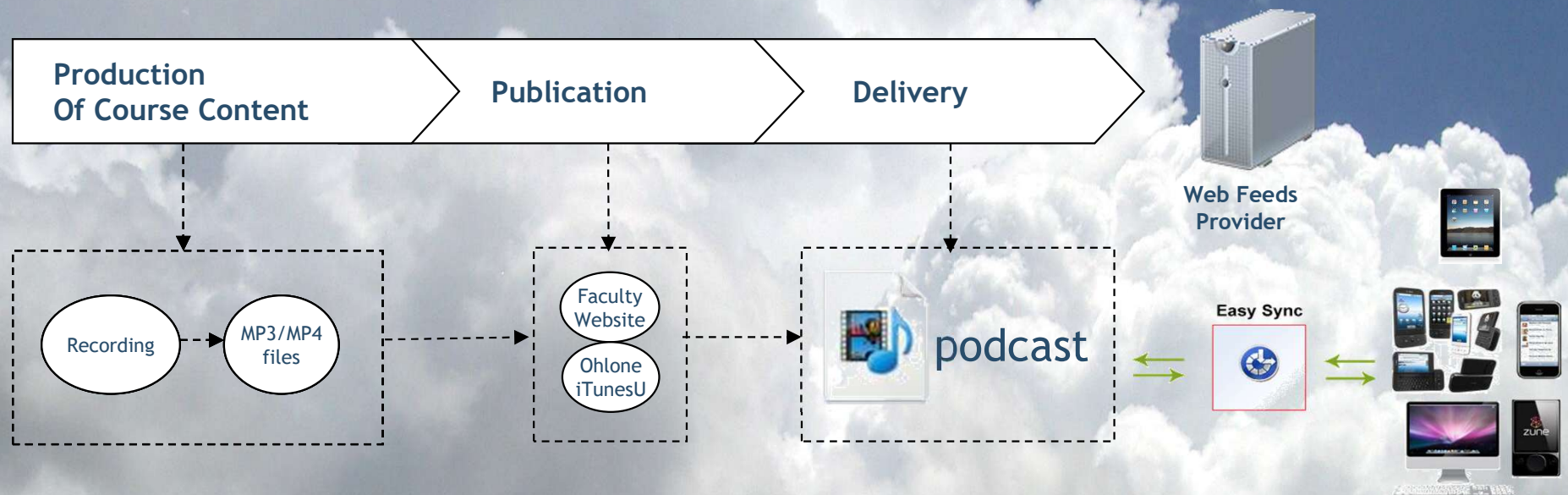


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Ohlone College's Podcasting Service



Internet





Ohlone College's iTunesU



Podcasting
reinforces
learning

Name	Time	Artist	Album	Price
1 The New England Way	1:17:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
2 Mercantilism & Enlightenment	53:15	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
3 Ben Franklin & Stage 1 of Rev	1:19:10	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
4 Ch 6/2 part 1: Road To Revolution 2/26	50:06	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free
5 Ch 6/2 part 2: Phases of Revolution 2/28	52:11	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free
6 Final Phases to Revolution	1:20:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
7 Revolutionary War	1:13:31	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
8 The Experiment Begins	1:20:29	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
9 President Washington	1:31:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
10 Jefferson's Revolutions	39:36	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
11 Early National Period	1:13:12	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
12 Manifest Destiny	1:04:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
13 American Exceptionalism	1:25:58	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
14 Economic Development North v South	1:24:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
15 War With Mexico	1:32:46	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
16 Prelude to Civil War	1:26:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
17 Civil War part 1	1:02:40	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
18 Civil War Part 2 + Reconstruction	1:27:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free
19 Reconstruction And Final exam Review	1:15:27	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free



Easy Sync



Ohlone College's iTunesU

<http://itunes.ohlone.edu>



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Ohlone Community College District - Community Demographics

Race	Fremont	%	Newark	%	Union City	%	District Total	%
American Indian	692	0.3%	29	0.1%	18	0.0%	739	0.2%
African American	6,012	2.9%	1,015	2.5%	4,195	6.0%	11,222	3.5%
Asian	84,657	40.6%	7,271	17.7%	20,919	30.0%	112,847	35.3%
Filipino	11,387	5.5%	3,058	7.4%	12,844	18.4%	27,989	8.8%
Hispanic	32,108	15.4%	14,656	35.6%	15,225	21.8%	61,989	19.4%
Pacific Islander	2,097	1.0%	564	1.4%	510	70.0%	3,171	1.0%
White	65,566	31.5%	13,223	32.2%	14,489	20.8%	93,278	29.2%
Other	5,936	2.8%	1,309	3.2%	1,548	2.2%	8,793	2.8%
Total	208,455		41,125		69,748		319,328	

Source: Ohlone College. (2009). *Ohlone data source: Institutional Research Office*



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Participants Grouped by Method of Classroom Instruction

Group	Type of instruction	Designation
1	Traditional classroom instruction	Control group
2	Classroom instruction supplemented with classroom podcast	Treatment group



Treatment Group

- During the Fall 2009 semester, all students from lecture-based course, History of the United States (HIST-117A) were invited to participate in the study.
- All students received traditional classroom instruction supplemented *with* classroom podcasts.
- The study sample of History of the United States for Fall 2009 semester was 72 students (or $N = 72$).



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Comparison Group

- The same lecture-based course from the Spring 2009 semester, taught by the same instructors in traditional classroom instruction without podcasting, was used as a control group.
- All students received traditional classroom instruction with *no* classroom podcast.
- The study sample of History of the United States for the Spring 2009 semester was 70 students (or $N = 70$).



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Data Collection Instrument

A three-part handout survey/questionnaire was used to collect students' responses regarding classroom podcast usage:

- (a) demographics questions
- (b) podcasting questions
- (c) open-ended questions





Demographic Profile

- Of the 72 participants, 85% ($n = 61$) responded to the in-class handout survey.
- 52.5% of the participating students were female.
- The age groups ranged from 19-or-younger (50.8%) through 20-24 years (41%) and 25-29 years (4.9%) to 40-or-older (3.2%).
- Whites had the highest participation rate, 42.6%, followed by Asians 19.7%, Hispanics, 11.5% , Multiracial, 11.5%, Filipinos 4.9%, and others 9.8%.
- Academic major include 19.7% in art, 23% in science, 16.4% in business, 6.6% in engineering, 3.3% in history, 8.2% undecided, and 23% other.



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Distribution of Ethnicity of the Sample

Ethnicity	<i>n</i>	%
Asian	12	19.7
Filipino	3	4.9
Hispanic	7	11.5
White, Non-Hispanic	26	42.6
Multi-racial	7	11.5
Middle Eastern	3	4.9
Hawaiian	2	3.3
Greek	1	1.6





Distribution of Year in College for the Sample

Year in college	<i>n</i>	%
First year	8	13.1
Second year	36	59.0
Third year	9	14.8
Fourth year	5	8.2
Fifth year or more	3	4.9



Distribution of the Academic Disciplines of the Sample

Academic major	<i>n</i>	%
Art	12	20.0
Business	10	23.3
Communication	4	16.7
Engineering	4	6.7
Science	14	15.0
Social Science	9	8.3
Undecided	5	6.7
Other	2	3.3



Distribution of Class Attendance by the Sample

Class attendance	<i>n</i>	%
Less than other classes	2	3.3
Same as other classes	42	70.0
More than other classes	16	26.7



Distribution of Listening Devices Reported by the Sample

Listening device	<i>n</i>	%
MP3/iPod player	10	16.7
Personal computer (PC)	32	53.3
MP3 player and PC	13	21.7
Cell phone/PDA, MP3, and PC	4	6.7
Cell phone/PDA and PC	1	1.7



Students' Attitudes Toward the Use of Classroom Podcasts

Response	Perceived ease of use (%)	Perceived usefulness (%)
Strongly Agree	32.8	18.0
Agree	44.3	45.9
Neutral	23.0	26.2
Disagree	0.0	4.9
Strongly Disagree	0.0	3.3



Results of the Study

- The results indicated *no statistically significant* differences between the average midterm and final examination test scores of the two groups.
- The average midterm and final examination test scores of the treatment group was *statistically significant*.
- The number of times students listened to classroom podcasts before the midterm was *statistically significant*.
- There was *no statistically significant* difference in classroom attendance.
- There was no *statistically significant* difference in the age groups.
- The difference was *statistically significant* in gender.



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Additional Findings

- *How could podcasts be improved to better meet your needs?*
- Written responses were assigned to four categories:
 - (a) quality of classroom podcasts,
 - (b) podcasting as a teaching and learning tool,
 - (c) learning preferences, and
 - (d) recommendations for improvement.



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Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Quality of classroom podcasts	<i>f</i>	%
Good the way it is	15	30
Classroom podcasts should be made available to the world	1	2
Did well on the midterm so don't need the podcast	1	2
Technological step in the right direction for students	1	2





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Podcasting as a teaching and learning tool	<i>f</i>	%
Helpful for reviewing class lectures	13	26
Helpful for reviewing interaction between teacher and students	1	2





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services ($N = 50$)

Learning preferences	f	%
Preferred classroom lectures	10	20
Preferred audio/video podcast	6	12
Preferred MP3 or compatible audio file format	3	6



Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services ($N = 50$)

Recommendations for improvement	f	%
Arrange the podcasts in chronological order	4	8
Provide video podcast for visual learners	3	6
Provide better sound quality	3	6
Remove extraneous material from the podcasts	3	6
Expand use of the podcasts to other classes	2	4



Distribution of Final Course Grades for the two groups

Final course grade *n* %
(Treatment Group)

A	22	30.6 (-3.8)
B	28	38.9 (+13.2)
C	17	23.6 (-6.4)
D	4	5.6 (-0.1)
F	1	1.4 (-2.9)

Final course grade *n* %
(Control Group)

A	24	34.3
B	18	25.7
C	21	30.0
D	4	5.7
F	3	4.3



Instructor Feedback



Positive aspects include:

- (a) students can re-listen to the lectures as often as they like,
- (b) students can listen at their leisure, and
- (c) students who are struggling with learning note-taking skills and English as Second Language (ESL) students can benefit by podcast use.

Disadvantages include:

- (a) students may rely on podcasting and not attend lectures,
- (b) students download classroom podcasts but do not actually listen to them.



Recommendations



- The recommendation for school leaders is to invest in state-of-the-art technology infrastructure and faculty and staff training.
- The recommendation for the people in the leadership positions is to invest in faculty and staff educational training.
- The recommendation for educators should consider technology-based teaching as the leader in student learning.

*Not just for the sake
of offering podcast*



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Suggestion for Future Research



- Replicate this study in other subject areas to test the overall effectiveness of classroom podcasts and determine longitudinal effects and understand students' attitudes toward podcasting.
- On the recommendation of the students, video podcasts is also needed for visual learners. Further study is needed to determine the impact of audio and video podcasts on visual learners.
- A book like device, the iPad (April, 3, 2010), which has the ability to serve for Web surfing, e-reader, game device, audio/video player, and multipurpose multimedia device.



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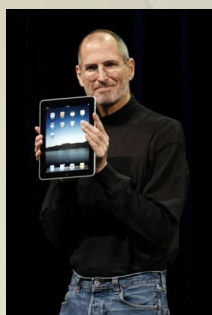
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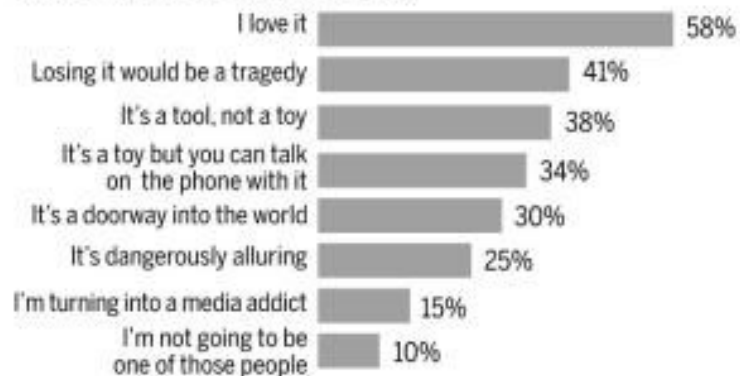


Is it a teaching and learning tool or a toy?



iPhone opinions

From a 2009 survey of 200 Stanford University undergraduates:
Do any of the following describe your own feelings about your iPhone? Please check all that apply:



Source: Professor Tanya Luhrmann, Stanford University

MERCURY NEWS



Professor Tanya Luhrmann, Stanford University



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Getting Started with Podcasting

- Digital recording device
- Laptop or PC
- RSS Server (or Web Feed Server)
- Subscribe to RSS Feeds



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Podcasting Equipments



It's your choice



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RSS Feed software Packages



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Conclusions



1. This study shows that podcasting has a potential to improve student learning, but it is not for every student.
2. The findings from this study can be used to help educators to determine the necessary balance between podcast learning and traditional classroom interaction.
3. The findings from this study could be used to expand technology-based learning and provide assistance to educators who desire to develop and share podcasts containing course materials with students.
4. Data gathered in this study could be used to provide leadership and support for future directions of technology-based instruction in higher education.
5. As this study showed that recurring training programs for faculty are essential to conduct podcasting (or technology-based) teaching and learning effectively.



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PodCast

Q & A



Thank you!



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Where To Find More Information



- iTunes RSS Feed Specification
<http://www.apple.com/itunes/podcasts/specs.html>
- RSS 2.0 Spec at Harvard Law
<http://cyber.law.harvard.edu/rss/rss.html>
- 3CRSS Media Solutions
<http://www.3cm mediasolutions.org/pages/solutions/index.html>
- RSS Viewer Applet for navigating and viewing RSS channels
<http://www.webreference.com/xml/column7/>
- RSS Feeds
<http://www.rss-specifications.com/rss-submission.htm>
- RSS Tools
<http://www.rss-tools.com>
- FeedForAll software allows users to create, edit, and publish RSS feeds
<http://www.feedforall.com>

